Guideline-adherence judgment workflow — rules decide; RAG supplies evidence and handles exceptions

A One verdict per patient × guideline × item (selection · dose · timing)

① Case card
only values reported before T+0
regimen R = agents started within 24 h
risk factors, comorbidities, severity inputs

② Compute conditions (code)
inpatient = ICU (all)
severe: ventilation · vasopressor · ≥3 minor criteria
MRSA risk: isolate ≤ 1 y
Pseudomonas risk: admission + IV abx ≤ 90 d
each condition = true / false / unknown

③ Find the matching row (rule table)
look up KR and INTL separately
no row → not applicable
'unknown' → evaluate every candidate row
same conclusion → adopt; differs → undeterminable

④ Compare regimen (sets)
R vs recommended S / alternative A
decision table in panel B
dose · route · timing are separate items

⑤ Aggregate
adherence = (recommended + alternative)
÷ (all − not applicable − undeterminable)
lower bound: undeterminable counted as non-adherent
report by group × guideline

Sample of 20 (+100 random)

LLM + RAG verdict
same card + 5 retrieved chunks
same 8 verdict values as JSON

Compare with rule verdict
agreement rate
disagreement types: LLM error /
condition outside the table / missing data

Clinicians (disagreements only)
confirm justified deviations
blind rating of both report types

B Regimen decision table — per patient, binary (example: severe CAP, no MRSA/Pseudomonas risk → S = ceftriaxone + azithromycin, A = levofloxacin)

R = S	**Adherent (recommended)** — e.g. ceftriaxone + azithromycin
R = A	**Adherent (alternative)** — e.g. levofloxacin alone · allergy data absent, so accepted without proof; share reported separately
R ⊃ S (recommended + extra agent)	**adherent if the extra agent is explained by another risk row, else non-adherent (excess)** — e.g. + vancomycin: adherent with MRSA history, excess without
R ⊂ S (only part of the recommendation)	**Non-adherent (missing component) — not 50%** — e.g. ceftriaxone alone (azithromycin missing). Component overlap only as a secondary metric
R ∩ S = ∅	**Non-adherent (different agents)** — e.g. cefepime + vancomycin without risk factors
no R · surgical cefazolin only · viral co-infection	**Not applicable (out of denominator, counted)** — e.g. cefazolin alone: 310 patients
an 'unknown' condition and candidate rows disagree	**Undeterminable (out of denominator; non-adherent in the lower bound)** — e.g. no vitals → severity unresolved (≈ 29%)
non-adherent but the card holds a reason	**Justified deviation (only with evidence cited by clinician / LLM)** — e.g. cefepime chosen for prior Pseudomonas isolate

C Pre-specified handling rules

- **card variable absent from the guideline (lactate, CRP …)** → not used in the verdict; context for the report only
- **guideline condition absent from the card (allergy, multilobar imaging)** → treated as 'unknown'; alternatives accepted as adherent without proof
- **two antibiotics given, only one in the guideline** → patient-level non-adherent (missing); never counted as 50%
- **broader than recommended** → adherent if the risk factor is in the card, else non-adherent (excess)
- **KR and INTL guidelines disagree** → judge each, report both; rule-table relation (common / conflict)
- **dose · route · timing** → separate items; mg/kg drugs undeterminable without weight
- **patients the guideline does not cover (prophylaxis, viral pneumonia)** → not applicable; counted and removed from the denominator
- **many undeterminable cases** → removed from the denominator, plus a lower-bound rate counting them as non-adherent
- **LLM verdict differs from the rule** → compared on the sample only; the adherence rate comes from rules alone

Adherence (selection) = (recommended + alternative) ÷ (all − not applicable − undeterminable) · also report: n not applicable, n undeterminable, non-adherence subtypes, alternative share, lower bound